Syllabus Year: 2024-2025

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2025

Bachelor of Computer Applications SEMESTER - I

Course Code: BCA103-1 **Max Marks: 50**

Course Name: PRINCIPLES OF SOFTWARE DEVELOPMENT - I **Duration: 02:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Design and Develop ethical software product.

CO2: Apply terms, concepts, and tools of relational database management systems.

CO3: Construct database projects using DDL and DML queries.

Section A **2*5=10**

1)	CO2: Explain the difference between a primary key and a foreign key in a relational database.	**2**	(CO2, L3)
2)	CO1: Define the principle of software development known as 'code review' and explain its importance in ethical software product design.	**2**	(CO1, L3)
3)	CO3: Describe the structure and purpose of a Data Dictionary in database management.	**2**	(CO3, L3)
4)	CO1: What are the ethical considerations a developer should keep in mind while designing software products?	**2**	(CO1, L3)
5)	CO2: List and briefly describe the four types of relational database normalization forms (1NF, 2NF, 3NF, BCNF).	**2**	(CO2, L4)

Section B **5*4=20**

6)	CO1: Part a) What is meant by 'modular design' in software development? Part b) How does modular design help in maintaining and updating software products?	**5**	(CO1, L4)
7)	CO3: Part a) Write an SQL DDL command to create a table named 'Employees' with fields for employee_id, name, and department. Part b) Write an SQL DML command to insert a new record into the 'Employees' table.	**5**	(CO3, L4)
8)	CO2: Part a) Explain the concept of 'data integrity' in relational databases and list two common data integrity constraints. Part b) Write an SQL DDL command to enforce referential integrity between two tables, 'Departments' and 'Employees'.	**5**	(CO2, L4)

Section C **10*2=20**

9)	CO3: Part a) Discuss the importance of using DDL and DML commands in database management. Part b) Write an SQL DDL command to create a view that shows all employees from the 'Sales' department.	**10**	(CO3, L5)

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10)	CO2: Part a) Explain the concept of 'transactions' in relational databases and their significance. Part b) Write an SQL command to start, commit, and rollback a transaction that involves updating employee salaries.	**10**	(CO2, L5)

Revised Bloom's Taxonomy (RBT) Levels

L1 - Remember
L2 - Understand
L3 - Apply
L4 - Analyse
L5 - Evaluate
L6 - Create

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